Thermal Science and Engineering Progress 42 (2023) 101868

Contents lists available at ScienceDirect

Thermal Science and Engineering Progress

journal homepage: www.elsevier.com/locate/tsep

Steam methane reforming over a preheated packed bed: Heat and mass transfer in a transient process

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ARTICLE INFO

Keywords:
Methane
Steam methane reforming
Preheated catalyst bed
Hydrogen-rich fuel
Numerical modelling

ABSTRACT

The primary method of large scale hydrogen production is steam methane reforming (SMR) in the packed bed reactors filled with the catalysts. In these reactors, heat is mostly supplied through the reactor wall to compensate for the endothermic effect of SMR reactions. In this paper, an alternative route of steam methane reforming process over a preheated packed bed filled with Ni-based catalyst particles was studied. Consequent packed bed heating with hot flue gas and cooling with H_2O - CH_4 endothermically reacting mixture was investigated. Numerical model has been developed for the cooling and heating processes and realized via Ansys Fluent. The analyses were performed for various steam-to-methane ratios and different initial temperature distributions. Results are indicating that the packed bed volume averaged temperature increases by 98 °C after 15 s of heating with flue gas. Packed bed cools down by more than 125 °C after 15 s of the SMR process. Mixture species distribution was also analysed along the reformer length. Hydrogen outlet molar fraction constantly decreases from 0.38 to 0.25 at the process beginning and end, respectively. The highest hydrogen production was observed on the half of the packed bed closer to the reformer outlet.

1. Introduction

The last decade has seen increasing fossil fuel consumption that lead to problems associated with climate change. The main problem of climate change is global warming which emerged due to extensive CO_2 emissions as a result of fossil fuel combustion. One of the promising ways to contain global warming is a transition to clean energy sources. Hydrogen production is a key technology in that transition [1–4]. Due to hydrogen CO_2-free combustion, it can substitute fossil fuels and thus be a bridging technology for the transition to clean energy sources. Therefore, technologies aimed at hydrogen generation will evolve dramatically. Constant growth in the world's hydrogen demand will lead to a hydrogen consumption of 200 Mt/year by 2030 [5]. Steam methane reforming (SMR) is a dominant technology for large-scale hydrogen production and accounts for about 80% of generated hydrogen [6].

Methane reforming is a process of methane transformation into a syngas fuel (with a high concentration of H_2) in the course of endothermic reactions. It is widely popular for many reasons: natural gas is abundant and cheap fuel [7], high process efficiency [8,9], low capital and production cost [10]. Moreover, methane reforming is an environmentally friendly technology as NO_x emissions are low and the process can reach net-zero emissions if carbon capture is utilized [11]. Due to these factors, methane reforming is a convenient hydrogen production technology and will be actively used in the coming years. However, methane reforming heavily depends on fossil fuel extraction and will be replaced by renewable hydrogen production technologies after these technologies will develop and become more economically feasible. [12,13].

Numerous articles were dedicated to SMR process improvement [14–19]. One of the main ways is to adjust the operating conditions, as SMR reactions heavily depend on temperature and pressure [20]. Peng and Jin [21] simulated the SMR process under various operating conditions: gas temperature (700 to 1100K), pressure (1 to 30 bar) and steam-to-methane ratio (1 to 5). They concluded that high temperature, low pressure and high steam-to-methane ratio are the most beneficial conditions for high hydrogen yield. Wachter et al. proposed to introduce methane partial oxidation along with steam reforming reactions. The main idea was to suppress coke formation as well as to lower the quantity of H_2O required for the reaction. Results show that the feed ratio of $CH_4/H_2O/O_2$ as 1/1/0.3, respectively, is the most desirable for high H_2 production and low coke formation. Combined SMR reactor modelling and whole H_2 production analysis were performed by Han et al. [22]. The authors conducted 3D CFD-modelling of the reformer with catalyst bed and whole H_2 production process simulation via Aspen Hysys. It was found that in some cases reactor efficiency increment does not

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https://doi.org/10.1016/j.tsep.2023.101868
Received 23 November 2022; Received in revised form 24 April 2023; Accepted 25 April 2023
Available online 2 May 2023
2451-9049/

always lead to increment in H_2 production so a new set of optimal operating conditions was obtained.

To enhance the reforming process efficiency it is important to study both the chemical equilibrium and chemical kinetics inside the reactor [23]. Equilibrium values are normally derived through thermodynamic study [24] and reaction kinetic values are usually obtained from experiments and CFD-modelling.

A thermodynamic analysis is a convenient instrument for determining the performance of different catalyst compositions. Under the different operating conditions, it is possible to reveal the maximum packed bed hydrogen production. Li and coworkers [25] investigated the performance of palygorskite supported nickel catalyst in the methane dry reforming process. They evaluated H_2 production and CH_4 conversion for 5 catalyst preparation methods under temperature range of 100-1200 °C and CH_4/CO_2 ratio of 0.5–1.5. Thermodynamic equilibrium was simulated in Aspen plus software. Results showed that the hydrothermal precipitation catalyst preparation method is the best among the 5 methods considered. A thermodynamic analysis is also used to assess different methods of methane reforming process. Three different hydrogen production technologies: autothermal methane reforming (AMR), dry methane reforming (DMR) and steam methane reforming (SMR) were compared by Carapelucci and Giordano [26]. The author's main comparison parameter was H_2 production efficiency. They found that DMR technology is worse than SMR due to the higher reaction temperature requirements for the same CH_4 conversion rate. But in the case of DMR operation under autothermal conditions the H_2 production rate increased by 9%, thus approaching the comparable with SMR values of efficiency.

As the researchers come up with new reactor designs they tend to use an experimental approach to collect the process data. Ma et al. proposed the use of electricity as the heat source for endothermic SMR reactions [27]. The novel electrified heat pipe reactor showed high-temperature uniformity (96.7%) across the conventional $Ni–Al_2O_3$ catalytic packed bed. Also, a high CH_4 conversion rate of 80% and a high H_2 volume fraction of 70% were achieved at a steam/carbon rate of 3 and heating temperature of 620 °C. The proposed reactor design had a much smaller volume than the conventional furnace SMR reactor [28].

Experimental investigations are traditionally bounded with difficulties such as high time requirements, large installation costs and the inability to view all process parameters due to measurement limitations [29]. In the last decade, CFD modelling becomes a prominent instrument for the study of the methane reforming process [30–35]. With the help of computer modelling, it is able to compare various design solutions, the influence of operating conditions and views all the temperature and species distribution both in steady and unsteady process. Wu and coauthors numerically studied the steam methane reforming over spherical catalysts with axial variable diameter [36]. The main finding was that H_2 production increases by 4.5% in structures with variable axial diameter due to the flow disturbances that enhance the mixing and, consequently, the reaction rate. Upadhyay et al. sought ways of improving hydrogen production in membrane reformers [37]. The authors decided to use computer modelling in the program Ansys for the determination of optimal operating conditions. Results indicated that temperature window 673–973K is optimal for reaction performance and hydrogen permeation. Reduction in mixture residence time led to H_2 production and permeation decline due to the limited reaction time. Also, a change in steam/carbon ratio from 1 to 3 led to a 4% increase in methane conversion.

The most common route for the SMR process is chemical reactions with external heat supply occurring in the vicinity of porous catalysts [38–40]. In these reactors, steam is premixed with natural gas and then supplied to the reaction zone. External heat for endothermic reactions could be supplied to the reactor walls via fossil fuel combustion or alternative routes could be chosen: solar heating [41], Joule heating [20] and waste heat recovery [42–45]. The reaction itself occurs on the packed bed with catalysts of various shapes and materials. However, that method of heat exchange has a considerable disadvantage bonded with the reformer wall material. Due to the high temperature of exhaust flue gases reactor wall has a high temperature from the heat supply zone and non-uniform temperature distribution from the reaction zone as strongly endothermic reactions of SMR are occurring [46]. Thus, a reactor wall is under considerable heat stress that may lead to material degradation. Moreover, heat exchange between the exhaust gases and reaction mixture is hindered by the thermal resistance imposed by the reactor wall.

Hisashi Kobayashi proposed a regenerative cycle for the methane reforming reactions [47]. The principal scheme is presented in Fig. 1. The whole process consists of consequent heating and cooling of the packed bed. At first, the catalyst packing in Reformer 1 is preheated by the exhaust flue gases to a temperature, that is favourable for steam methane reforming reactions. After the packing is heated enough the flue gases are redirected to the Reformer 2. Meanwhile, the reaction mixture of CH_4 and H_2O is fed to the hot Reformer 1, where the accumulated heat is consumed in the course of steam methane reforming reactions. After reformer 1 is cooled to a certain temperature at which the reaction rate is considerably slowed, the flows are redirected and Reformer 1 starts heating by the flue gases, while Reformer 2 starts cooling by the endothermic reactions. Reaction products — gas with high H_2 concentration are usually stored or used immediately as a burner fuel [48]. The main advantage of this heat supply method is the absence of the reactor wall's thermal resistance. Previously, the chemical kinetics on preheated Ni-based catalyst bed was numerically investigated [49]. The 2D CFD model was developed in Ansys Fluent software, and pressure drop, species mole and mass fractions were analysed. However, process visualization could be improved as calculations were carried out in steady-state formulation. The unsteady formulation allows for deep analysis of the temperature and species distribution in different moments of the cycles. During the packing cooling heat from the catalyst bed is consumed at a certain rate until the temperature of catalysts reaches a value that is not favourable for SMR reactions. Then the process should be switched from reaction mode (endothermic reactions occur) to the heating mode (packed bed heating). Catalyst temperature and species distribution are constantly changing over time. Therefore, it is important to study the process in unsteady formulation to capture the changes in catalyst temperature and species distribution.

Transient steam methane reforming reactions have captured the attention of scientists. Sheu et al. used transient calculations for insight into the sorption-enhanced steam methane reforming reaction kinetics [50]. CaO sorbent was used for CO_2 sorption and heat flux was supplied through the reactor wall. Authors found that as the reaction time increases the maximum conversion rate and temperature moves away from the wall towards the central region. Moreover, they observed the formation of the CO_2 sorption reaction peak and its consequent shift downstream at a practically constant speed. The transient process in an industrial-scale steam methane reforming reactor was modelled by Wang et al. [51] to obtain the changing pattern of CH_4 conversion. Heat for endothermic reactions was supplied through the wall, therefore, the highest methane conversion was observed near the wall and in the centre parts of the catalyst bed, CH_4 reforming was considerably lower. During the first 40 s of reforming the H_2 concentration near the reactor wall was growing and the H_2 mole fraction in the bulk of the bed began to increase only after about 100 s. That dependence illustrates slow heat transfer to the catalyst, therefore it takes considerable time to heat the bulk of the bed. Yun and Yu [52] investigated the transient behaviour of a low-temperature steam methane reformer and found out that among all the parameters, required hydrogen production is the most influential factor. Transient simulations of steam methane reformer heated by concentrated solar irradiation were performed by Wang and coauthors [53]. Their focus was mainly on temperature and species distribution change over time.

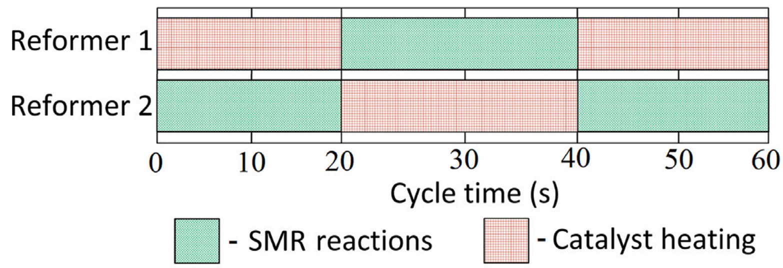

Fig. 1. Principal scheme of reformers, working by a regenerative scheme [47].

Mixture temperature and H_2 production steadily increase until the process reaches equilibrium values in 250 s.

Recent articles are focused mainly on SMR reactions with heat supply through the reactor wall. However, the preheating of the catalyst for the steam methane reforming process has considerable advantages: the absence of the thermal resistance imposed by the reactor wall and lesser thermal stress that the reactor wall experiences. Also, a mixture of methane and steam has an improved pattern of heating as porous catalyst is heated more uniformly. For comparison, in the conventional process (heat flux through the reactor wall) the central parts of the packed bed are poorly heated and CH_4 conversion is hindered in that region [51]. Moreover, the use of flue gas in a regenerative heating hinders carbon deposition as flue gases contain considerable amount of water vapour [54]. High water vapour content leads to reaction with the carbon species on the catalyst surface, leading to their oxidation and removal. This means that even less carbon will accumulate on the catalyst surface.

Therefore, SMR reactions over a preheated nickel catalysts were investigated in this article. Numerical simulations in Ansys Fluent were completed to calculate and visualize the temperature fields, species distributions and cycle time both for a cycle with SMR reactions and the consequent cycle of catalyst heating by the flue gases. The temperature decrease and increase in reactor over time were predicted for cooling (with SMR) and heating cycle, respectively. Mixture species distribution was observed along the reactor length at different flow time. Also, length of the cooling and heating cycle was obtained and compared. The numerical calculations were carried out for H_2O/CH_4 ratios (β) of 1.0, 2.0 to analyse the process variables under different operating conditions.

2. Methodology

2.1. Computational geometry and meshing

In this study, a steam methane reforming reactor operation was numerically modelled. The first step of creating a numerical model is building the computational domain. In the present simulation steam methane reforming reactor geometry (Fig. 2) was simplified to a small cylindrical tube with 10 cylindrical catalyst particles inside. This geometry simplification is widely used to lower the computational time [55, 56]. The reduction in computational time is necessary for the present model as simultaneously several models were used: turbulent flow, heat transfer, flow in porous media, species transport with user-defined function for chemical reaction kinetics. Conjugate solution of these models in a transient state requires significant computational time. The reduced in size packing bed will not significantly affect the simulation results as reformer wall is adiabatic and SMR reactions occur only inside the catalyst volume. The reformer tube diameter is 14 mm to allow the unstructured packing of cylinders with a height and diameter of 10 mm. Space before the packing bed was created to obtain the developed flow. The outflow zone after the packing bed was inserted to ensure the dissipation of possible pressure fluctuations that may cause the reversed flow. Also, catalysts are considered a porous media and, therefore, they create additional resistance to the gas flow, which is described in the next section.

The next step after geometry creating is mesh generation. The computation domain was divided into triangle cell with total number of 7.9 million elements. Mesh is presented in Fig. 3. One of the major problems with mesh in packed bed simulations is contact point treatment. On one hand, contact points between particles and particle–wall contacts require a very refined mesh to correctly capture the flow phenomena. On the other, these additionally refined zones lead to a considerable mesh size gradient, which creates numerical errors [57]. To solve that problem several techniques were created, the most common are bridging and scaling [58]. In the present numerical model, a scaling method was applied, i.e. the cylinder volume was shrunk to 0,99 of the initial volume. In packed beds with cylindrical catalysts, the turbulent fluctuations occur rather often [59]. To capture the turbulent flow in the near-wall region correctly the inflation procedure was used. The inflation layer consists of 5 hexahedral layers and is shown in Fig. 3. The first layer thickness was 1.2×10^{-5} m with a growth rate of 1.2, so the total boundary layer thickness was 8.9×10^{-5} m. The first cell height was calculated according to the dimensionless distance from the surface to the first grid node y^+ so the expression: $y^+ \sim 1$ remained true.

2.2. Governing equations and reaction kinetics

To study the distribution of reaction species and temperature along the reformer volume the simulation in Ansys Fluent 19R2 software was performed. The model has the following assumptions: the process is unsteady, adiabatic conditions are imposed on the reactor wall, catalyst thermal conductivity remains constant [60], the fluid is deemed incompressible and radiation heat transfer is neglected. The reactor walls are deemed adiabatic as high-quality insulation should cover them [61], however, the effect of the heat losses through a reactor wall was not addressed in the current investigation. A set of differential equations was solved for every mesh cell to obtain the computational results.

The continuity equation in unsteady formulation is written as:

$$\nabla \cdot (\rho \vec{v}) + \frac{\partial \rho}{\partial t} = 0, \tag{1}$$

where ρ represents fluid density, $\vec{v}$ is a flow velocity.

Momentum equation is expressed [62]:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = \nabla \cdot (\bar{\bar{\tau}}) - \nabla \cdot p + \rho \vec{g} + \vec{F} + S_i, \tag{2}$$

where the additional source term S_i is added to take the effect of packed bed resistance into account, $\vec{F}$ is external body force, $\vec{g}$ represents gravitational body force, $\bar{\bar{\tau}}$ is stress tensor and p – the static pressure.

The stress tensor $\bar{\bar{\tau}}$ could be described as:

$$\bar{\bar{\tau}} = \mu(\vec{v}^{\mathrm{T}} + \nabla \vec{v}) - \frac{2}{3}\mu \nabla \cdot \vec{v} I, \tag{3}$$

where μ represents molecular viscosity and I is a unit tensor, the far right term accounts for the effect of volume dilatation.

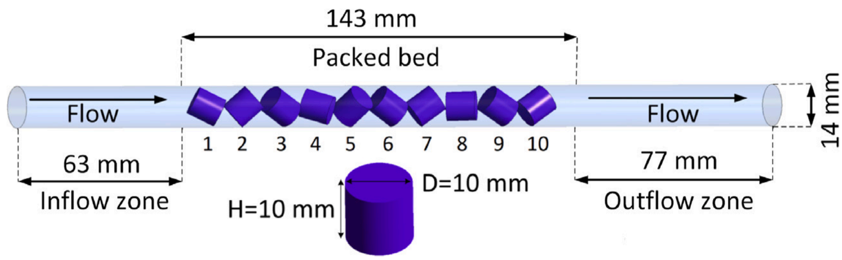

Fig. 2. The computational domain: reformer filled with 10 cylindrical particles.

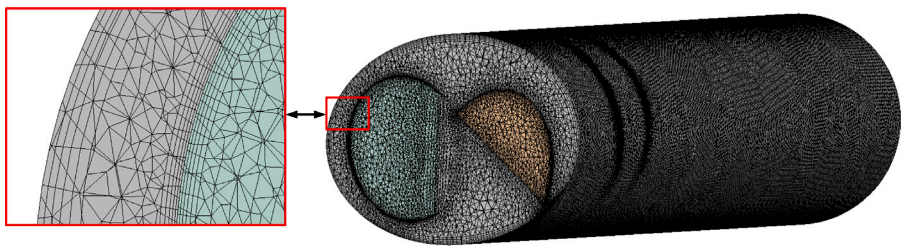

Fig. 3. The mesh of a reformer filled with cylindrical catalysts.

The source term Si in Eq. (2) is comprised of viscous and inertial loss terms:

$$S_i = -v_i \cdot \left(\frac{\mu}{\alpha} + C_2 \frac{1}{2}\rho|v|\right), \tag{4}$$

where α and C_2 are permeability and internal drag coefficient, respectively.

Gas flow through the packed bed was considered turbulent due to velocity fluctuations near the catalyst and reactor surface. Therefore, gas flow through a porous medium of catalyst particle was set by the Ergun equation written below [18]:

$$\frac{\Delta p}{L} = \frac{150\mu}{d_p^2}\frac{(1-\gamma)^2}{\gamma^3}v + \frac{1.75\rho}{d_p}\frac{(1-\gamma)}{\gamma^3}v^2, \tag{5}$$

where d_p is particle pore mean diameter, γ is particle porosity, L represents the packed bed length.

To obtain the values of inertial and drag resistance, the Ergun equation was used so the coefficients were found according to [63]:

$$\alpha = \frac{d_p^2}{150}\frac{\gamma^3}{(1-\gamma)^2}; \tag{6}$$

and

$$C_2 = \frac{3.5}{d_p}\frac{(1-\gamma)}{\gamma^3}. \tag{7}$$

Energy equation was modified as porous media region is present so for the gas phase elements and solid region it takes following form:

$$\frac{\partial}{\partial t}(\gamma \cdot \rho E_f + (1-\gamma)\rho_s E_s) + \nabla \cdot (\vec{v}(\rho E_f + p)) = \nabla \cdot \left[\nabla T \cdot k_{\text{eff}} - \left(\sum_i h_i \vec{J}_i\right) + \bar{\bar{\tau}} \cdot \vec{v}\right] + S_r^h, \tag{8}$$

where E_f and E_s are gas mixture and solid medium energy, respectively; ρ_s is solid medium porosity, h_i is enthalpy of mixture element i, J_i represents diffusion flux of element i, S_r^h is a source term accounting for the enthalpy change due to occurring reactions.

Effective thermal conductivity k_{eff} is calculated as a volume average of fluid and solid phase thermal conductivity:

$$k_{\text{eff}} = \gamma \cdot k_f + (1-\gamma) \cdot k_s, \tag{9}$$

where k_f and k_s are fluid and solid phase thermal conductivity, respectively.

The local mass fraction of each element Y_i in gas mixture is derived from:

$$\frac{\partial}{\partial t}(\rho Y_i) + \nabla \cdot (\rho \vec{v} Y_i) = -\nabla \cdot \vec{J}_i + R_i, \tag{10}$$

where R_i is the net rate of production and consumption of species i in the course of chemical reactions, $\vec{J}_i$ represents diffusion flux of species i, Y_i - mass fraction of element i.

Mass diffusion flux of element i in turbulent flow has following form:

$$\vec{J}_i = -\left(\rho D_i \cdot \nabla Y_i + \frac{\mu_t \cdot \nabla Y_i}{\text{Sc}_t} + D_{i,T}\frac{\nabla T}{T}\right), \tag{11}$$

where D_i and $D_{i,T}$ are mass and thermal diffusion coefficients for mixture species i, respectively; Sc_t represents turbulent Schmidt number and μ_t — turbulent viscosity, respectively.

Standard Shear Transport (SST) $k-\omega$ model was chosen to predict the turbulent flow. That model combines the benefits of the $k-\epsilon$ model in free stream and $k-\omega$ model in the near-wall region and is recommended to use with chemical reactions [64]. The transport equations are written as:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j}\left(\Gamma_k \frac{\partial k}{\partial x_j}\right) + G_k - Y_k; \tag{12}$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i}(\rho \omega v_i) = \frac{\partial}{\partial x_j}\left(\Gamma_\omega \frac{\partial \omega}{\partial x_j}\right) + G_\omega - Y_\omega + D_\omega, \tag{13}$$

where Γ_ω and Γ_k represent the effective diffusivity of ω and k, respectively; G_ω defines the generation of ω, G_k represents turbulent kinetic energy generation due to mean velocity gradients, Y_ω and Y_k represent dissipation of ω and k due to turbulence, respectively; D_ω represents cross-diffusion term.

Cross-diffusion term D_ω is used to bind $k-\epsilon$ and $k-\omega$ standard turbulence model equations and thus build SST $k-\omega$ model equation. It has the following form:

$$D_\omega = 2\left(1 - F_1\right)\rho\sigma_{\omega,2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}, \tag{14}$$

where F_1 is a blending function and $\sigma_{\omega,2} = 1.168$.

Steam methane reforming is a multi-reaction process, among which 3 are the most important [65]: steam methane reforming reaction, water gas shift reaction and global steam methane reforming reaction:

$$CH_4 + H_2O \Leftrightarrow CO + 3H_2 + 206.1\ \text{kJ}; \tag{15}$$

Table 1
Catalyst properties.

Tube to particle diameter	–	1.41
Tube length	m	0.283
Number of particles	–	10
Catalyst particle diameter	m	0.1
Catalyst particle height	m	0.1
Particle surface area	m^2/kg_{cat}	4.5
NiO content	%	14.8
Effective thermal conductivity	W/m K	1
Specific heat	J/kg K	1000
Density	kg/m^3	1900
Particle void fraction	–	0.44

$$CO + H_2O \Leftrightarrow CO_2 + H_2 - 41.1\ \text{kJ}; \tag{16}$$

$$CH_4 + 2H_2O \Leftrightarrow CO_2 + 4H_2 + 165.0\ \text{kJ}. \tag{17}$$

To calculate the reaction rate of each reaction a comprehensively developed reaction kinetics for Ni-based catalysts was utilized [66]. The following equations were included in model to describe the volume reaction rates of reactions (15)–(17), respectively:

$$r_1 = \frac{k_1}{p_{H_2}^{2.5}}\left(p_{CH_4}\cdot p_{H_2O} - \frac{p_{H_2}^3\cdot p_{CO}}{K_{e1}}\right)\cdot\frac{1}{Q_r^2}; \tag{18}$$

$$r_2 = \frac{k_2}{p_{H_2}}\left(p_{CO}\cdot p_{H_2O} - \frac{p_{H_2}\cdot p_{CO_2}}{K_{e2}}\right)\cdot\frac{1}{Q_r^2}; \tag{19}$$

$$r_3 = \frac{k_3}{p_{H_2}^{3.5}}\left(p_{CH_4}\cdot p_{H_2O}^2 - \frac{p_{H_2}^4\cdot p_{CO_2}}{K_{e3}}\right)\cdot\frac{1}{Q_r^2}; \tag{20}$$

$$Q_r = 1 + K_{CO}\cdot p_{CO} + K_{H_2}\cdot p_{H_2} + K_{CH_4}\cdot p_{CH_4} + \frac{K_{H_2O}\cdot p_{H_2O}}{p_{H_2}}, \tag{21}$$

where p is gas partial pressure, $k_{1,2,3}$ is reaction kinetic rate constant, $K_{e1,2,3}$ is reaction equilibrium constant, K_{CO,H_2,CH_4,CO_2} is adsorption constant. All constants in Eqs. (18)–(21) are taken from Xu and Froment's study [66].

The Eqs. (19)–(21) were obtained by Xu and Froment for no diffusion limitation case for the catalyst particles of very small size (about 0.1–0.2 mm). To take into account an inparticle diffusion, the efficiency factors were involved to the equations for the reaction rate of consumption or appearing of each reacting spaces. For example, to estimate the methane consumption rate and carbon dioxide production the following equation is used:

$$R_{CH_4} = \eta_1 r_1 + \eta_3 r_3; \tag{22}$$

$$R_{CO_2} = -(\eta_2 r_2 + \eta_3 r_3); \tag{23}$$

where η_i is an efficiency factor for ith reaction, respectively.

2.3. Boundary conditions

In the present paper, steam methane reformer operation in a regenerative mode on a preheated nickel-based catalyst is studied. Catalyst affects the chemical reaction kinetic rate and adsorption constants significantly [67]. Therefore, its properties should be carefully chosen. The properties of the catalyst are listed in Table 1.

Reactor regenerative operation is consequent cycles of packed bed heating and cooling. In the cooling process, the heated to 927 °C catalyst particles are cooled with the reaction mixture flow (methane and steam) with an initial temperature of 426 °C. Heat stored inside the catalysts is consumed due to endothermic SMR reactions ((15) and (17)) and on the mixture heating. As a result, the catalysts are cooled down to a temperature that is far less favourable for SMR reactions. To close the cycle the catalyst heat should be replenished. In the present study flue gases with an initial temperature of 977 °C and composition listed in Table 2 are used for the catalyst heating. The initial catalyst temperature in the heating cycle is the average catalyst temperature at the end of the cooling cycle. The duration and species distributions in the aforementioned processes are of interest.

The boundary conditions are presented in the Table 2

For pressure–velocity coupling, the SIMPLE algorithm was used. Second-order discretization was applied for advection terms. Convergence criteria for energy equations was set to $1\cdot10^{-6}$, for other equations convergence criteria was $1\cdot10^{-3}$. To enhance solver stability under-relaxation factors were reduced by 0.05 for reaction species. Mixture specific heat capacity was calculated from individual species specific heat capacities according to mixing law. The specific heat capacity of species was written in piecewise polynomial form according to Yuan et al. [63]. The time step was set to 0.001 s so the Courant number (Cr) met the condition $Cr \ll 1$. The simulation time for the cooling cycle (with SMR) was 15 s and the catalyst heating cycle ended when the average catalyst bed temperature reached 927 °C, which is the cooling cycle's initial temperature. The operating temperature of 927 °C is encountered in high temperature SMR in gas turbine power plants [68]. Three cycles of consequent heating and cooling were simulated and analysed.

3. Results

In this section, CFD-results are presented. The results are divided into three sections: the SMR endothermic reactions over a preheated catalyst, catalyst heating with the flue gas heat and the analysis of consecutive cooling and heating cycles.

In this paper, the chemical kinetics for the steam methane reforming process were taken from the literature [66]. To set this kinetic model in Ansys Fluent solver, a user defined function (UDF) was prepared and compiled. CFD code was verified with the experimental data and analytical calculations in the authors previous works [30,56].

3.1. SMR reactions over preheated catalyst

The principle of preheated packed bed operation is to store and then provide the heat for endothermic reactions. The process of catalyst cooling occurs in an unsteady regime as heat is gradually consumed by the reaction mixture. Therefore, reaction kinetics in unsteady form was utilized to understand the temperature and species distributions over preheated Ni-based catalysts. The temperature distribution of the packed bed at several time steps: 0 s, 1 s, 2 s, 5 s, 10 s and 15 s are shown in Fig. 4. At 0 s the catalyst pellets have a uniform temperature of 927 °C. That temperature is obtained by the preheating of the pellets with hot flue gas. The assumption was made about uniform temperature on every catalyst. In real process the initial catalyst temperature may slightly differ. After the preheating the reaction mixture of CH_4 and H_2O is introduced. The stored in the catalysts heat is firstly consumed by the fluid on its heating and after the mixture is heated to 600 °C the active endothermic SMR reaction consumes additional heat [69].

The most important reactant is CH_4 (main fuel component) and the reaction product is H_2. To visualize the species distributions inside the catalyst pellets contours of methane and hydrogen mole fraction are presented in Fig. 5. In the initial moment (t = 0 s) species molar concentration is equal to zero as no mixture is flowing. In each consequent timestep, the methane concentration gradually decreases and the hydrogen molar fraction constantly increases. Higher hydrogen production is observed on the catalysts closer to the reformer exit. The bulk of methane conversion occurs in the catalyst centre as flow velocity there is quite low. Flow is hindered inside the catalyst due to the high inertial and drag resistance of porous media. However, mixture velocity is higher in the particle near the surface region and lower in its centre. The fluid that enters the inner region of the catalyst moves primarily due to the molecular diffusion, while the near wall catalyst

Table 2
Boundary conditions.

	Inlet temperature	Inlet velocity	Y_{CH_4}	Y_{H_2O}	Y_{CO_2}	Y_{N_2}	Pressure	Catalyst temperature
Units	°C	m/s	(–)	(–)	(–)	(–)	bar	°C
Reaction mixture	426	1.5	0.31–0.471	0.529–0.69	0	0	1	927
Flue gas	927	8	0	0.19	0.095	0.715	1	from cooling cycle

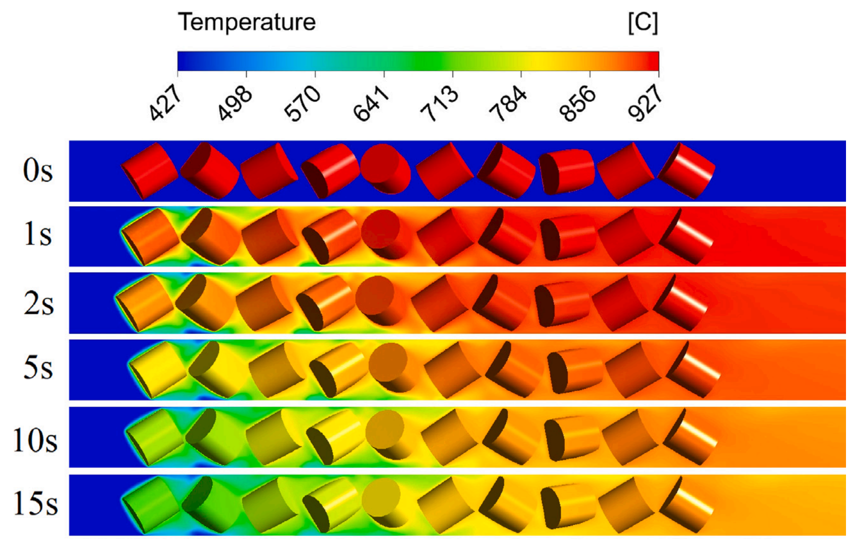

Fig. 4. Packed bed temperature distribution at different flow time.

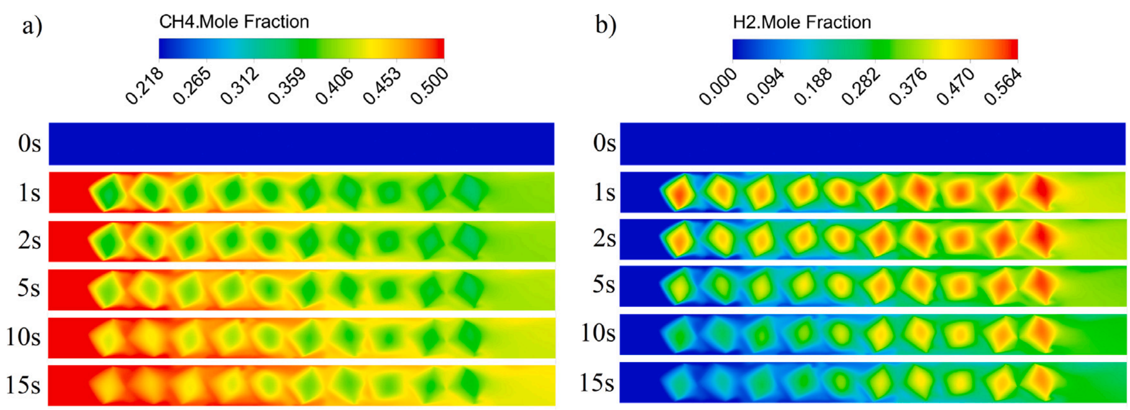

Fig. 5. Contours of (a) CH_4 and (b) H_2 molar fraction distributions at various time steps.

region is affected by the higher mixture velocity outside the catalyst particle [70]. Low flow velocity provides high residence time that is very favourable for the reaction rate [39].

The reaction products and residual reactant species average molar fraction was investigated on the reactor outlet. The dependence of the average species molar fraction on the flow time is depicted in Fig. 6. The molar concentration of reactants CH_4 and H_2O constantly increases in 15 s from 0.311 to 0.375 and from 0.217 to 0.313, respectively. Molar fraction of reaction products H_2 and CO_2 gradually decreases from 0.377 to 0.249 and from 0.094 to 0.062, respectively. The molar concentration of reactants and reaction products increases and decreases, respectively from the flow time of 0 s to 15 s. That is explained by the reaction rate decrease due to a decrease in catalysts temperature over time and therefore heat available for the reaction is constantly decreasing. The depicted in Fig. 6 dependence is practically linear with the more steep change of the species concentrations in the early flow time.

The change of catalyst volume average temperature over time is presented in Fig. 7, where the far left catalyst is №1 as in Fig. 2. The initial temperature of 927 °C was the same for all particles. Catalyst №2 was second to meet the reaction mixture, therefore it participated both in mixture heating and SMR reactions. At the first time steps (up to 7 s) cooling on the catalyst №2 is occurring by a quadratic dependence and then the behaviour is practically linear. As reaction rate dependency on temperature is non-linear [71], it can be said that quadratic temperature decrease is closely related to a quadratic decrease in reaction rate. Fig. 5 demonstrates a sharp reaction rate reduction on time, especially for the first particles. Particles closer to the inlet have lesser temperature than the ones down the stream. However, from catalyst №6 temperature difference is far less pronounced. Mixture temperature varies also insignificantly after the flow reaches catalyst №6. The biggest temperature differences between catalysts №2, 4 and 6 are caused by the active heat exchange with the reaction mixture. The further catalyst is from the reactor inlet, the lesser it participates

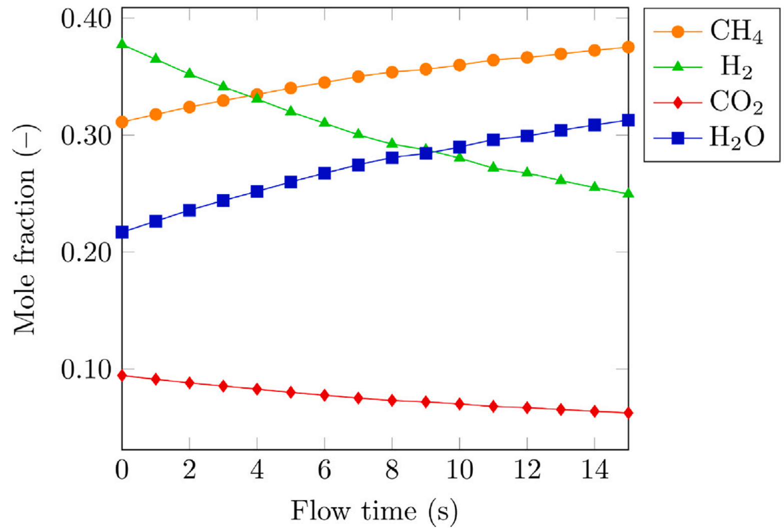

Fig. 6. Average species molar fraction on the reformer outlet at different flow time.

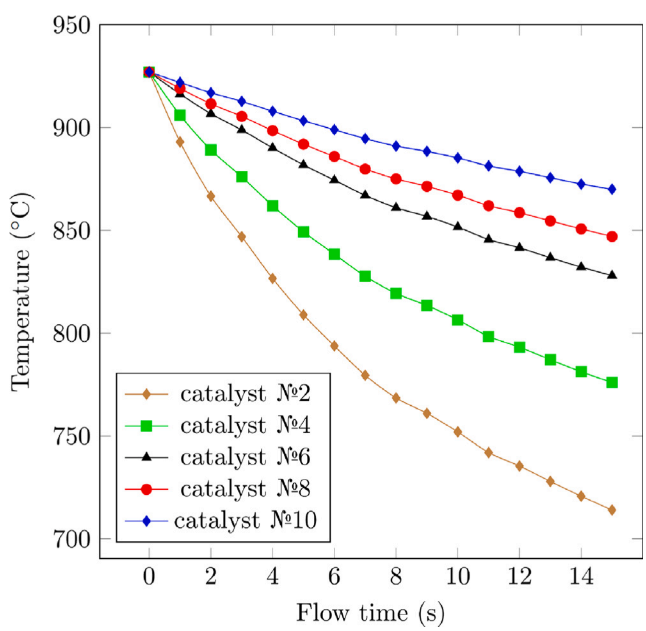

Fig. 7. Volume average catalyst temperature distribution at various time steps.

in mixture heating and, thus, a bigger share of stored heat is consumed due to the endothermic reactions.

3.2. Packed bed heating

Catalyst heating with flue gas heat is used to replenish the heat, that is consumed in the cooling cycle by endothermic reactions. The initial condition for temperature was taken from the end of the previous cycle (15 s) when the packed bed was cooled with SMR reaction. In other words, the average volume temperature of each catalyst at the end of the cooling cycle was set as the initial temperature for the heating cycle. The temperature field in the packed bed heating process is shown in Fig. 8. The inlet temperature of hot flue gas is 977 °C and the velocity is 8 m/s. The heating occurs gradually as the temperature slowly increases from the first (left) to the last catalyst particles (right). The average gas temperature at the reformer outlet increases from 785 °C to 911 °C. The process was completed at 21.8 s when the volume average temperature of all particles reached 927 °C.

The numerical values of volume-averaged particle temperature are presented in Fig. 9. The low temperature values on catalysts №2, 4 (as in Fig. 2) are due to the extensive heat exchange in the previous (SMR) cycle. The catalysts №2, 4 are heating constantly by a curved line from 714 °C and 776 °C to 959 °C and 945 °C, respectively. The other particles are cooled down in the early flow time and then the heating begins. At the beginning of the heating cycle, the bulk of the flue gas heat is consumed by the catalyst №1–4 so the flue gas temperature drops significantly. When the flue gas that is cooled down on the first particles reaches catalysts down the flow, catalyst cooling occurs as their temperature is higher than that of the flue gas. That behaviour is shown in Fig. 9 as the temperature of particles №6–10 is lower at flow time 2 s than at the beginning of the process (0 s). Also, it is observed that the closer catalyst is to the reformer outlet, the later its heating occurs. The temperature of practically all the pellets is increasing slower at the end of the heating cycle, especially for catalysts №1–5 that meet the flow first. Thus the heat transfer is hindered at the end of the cycle as it heavily depends on temperature difference [72]. The temperature difference between the flue gas and particles decreases as the heating of catalysts occurs and in the cycle end difference in particle to particle temperature is less pronounced than at flow time 0 s.

3.3. Consequent heating and cooling cycles

The temperature distribution on the catalysts at the end of the heating cycle as shown in Fig. 9 is different from initial conditions in the cooling cycle as shown in Fig. 7. The initial condition in the first cooling cycle was a volume average temperature of 927 °C on every catalyst. The volume average temperature at the end of the heating cycle on each catalyst was between 892 °C and 959 °C. Therefore it is important to investigate the SMR reactions in several consequent cycles of cooling and heating as temperature distribution influences reaction rates and heat exchange significantly [73].

In Fig. 10 volume average temperatures of all catalysts at different flow times under various H_2O/CH_4 (β) ratios for three consequent cycles of packed bed cooling and heating are presented. The range of $\beta$for the current investigation was chosen according to the common value encountered in applications for thermochemical recuperation systems [74]. In all cooling cycles (with SMR reaction) the average catalyst temperature at the time step 0 s is 927 °C but the temperature at the end of each cycle (15 s) is different. For the $\beta = 1$ (Fig. 10a) cycle 1 ends at a higher temperature than cycle 2, but at lower than cycle 3. In general, in the first cycle packing cooling is the strongest

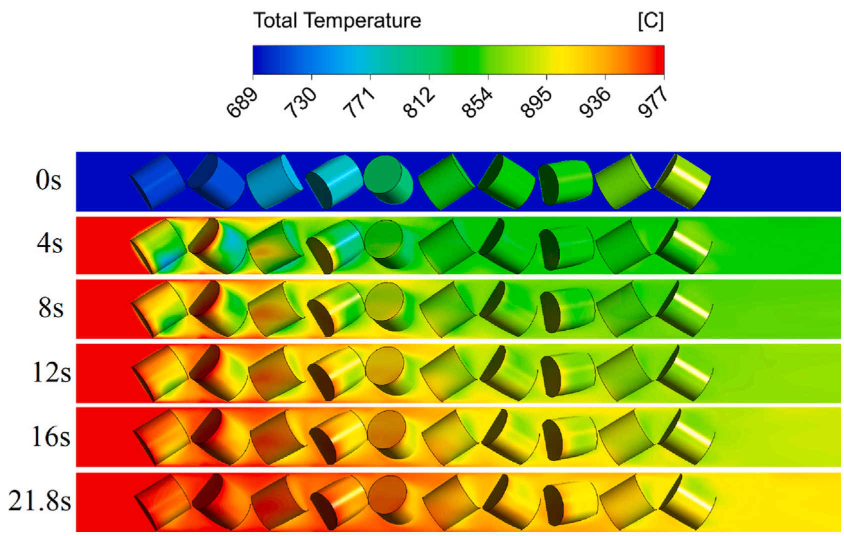

Fig. 8. Temperature contour in the packed bed heating process at different flow time.

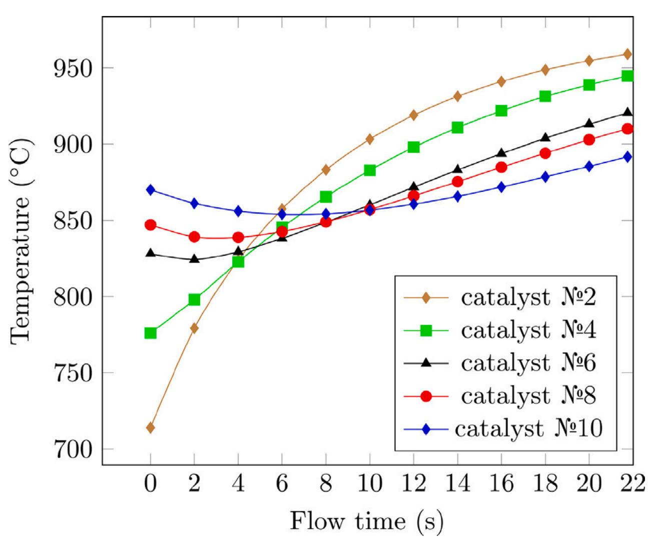

Fig. 9. Volume average temperature of catalysts №2, 4, 6, 8, 10 at different time steps in the heating cycle.

due to the uniform temperature of all particles. That high uniform temperature allows for all particles to participate in chemical reactions with a high reaction rate. Each heating cycle follows the cooling cycle, for example, after the cooling cycle 1 the heating cycle 1 occurs and so on. Average time for heating the packed bed to the volume average temperature of 927 °C was 21 s and varied from cycle to cycle. Packing heating with hot flue gases occurs by a curved line due to the temperature gradient reduction between flow and particles and therefore heat transfer coefficient decreases. The packed bed is heated to an average temperature of 927 °C after approximately 21 s and cooling cycles are completed after 15 s. So the packed bed heating was occurring on average at a 1.5 times slower rate, than its cooling. That effect is explained by the absence of an endothermic reaction, that intensifies the heat consumption in the cooling cycle [75]. Moreover, flue gas and particles have a lower temperature difference than reaction mixture and particles. For the steam-to-methane ratio of 2 (Fig. 10b) the heating time in the first cycle is similar to the graph with steam-to-methane ratio of 1 (Fig. 10a). However, the second and third heating cycles are finished in about 20 s, which is 2 s shorter. That behaviour is caused by the initial average temperature of all catalysts, which is considerably higher.

The temperature difference between cycles on every catalyst with the β ratio of 1 and 2 is shown in Fig. 11. Volume average temperature was measured on every catalyst at the end of each cooling cycle (with SMR reactions) and then the values at the end of the next cooling cycle were subtracted. The difference between the first and second cycles under $\beta = 1$ shows that the particle temperature during the second cycle is higher on the first half of the packed bed and lower on the half closer to the reformer outlet. That behaviour is due to the packing heating as its end temperature on the first catalysts is higher than 927 °C and the last catalysts have a temperature lower than 927 °C (Fig. 9), which is the initial temperature at the start of the first cooling cycle. At the end of the third cooling cycle, all catalyst temperatures are higher, which indicates that the reaction rates are decreasing and the packed bed becomes less effective. This effect is caused mainly by the lesser temperatures on the last catalysts at the start of the third cooling cycle because they are not able to provide enough heat for the same reaction rates as in the first cooling cycle, while the most heated catalysts (№1–5) are used primarily for mixture heating. For a mixture with ratio $\beta = 2$ a more pronounced temperature difference between cycles 1 and 2 is observed. After cycle 2 is completed the process reaches its fully developed state as the temperature difference between cycles 2 and 3 is minimal.

4. Conclusion

This work presents a temperature and species distribution analysis of the steam methane reforming (SMR) process over a preheated packed bed for an unsteady regime. The chemical kinetics of the process, temperature dependence of packed bed cooling and heating on cycle time with respect to changing steam-to-methane ratio are investigated.

Catalyst temperature for steam to methane ratio $\beta = 1$ in unsteady SMR process decreased rapidly throughout the whole packing bed due to the active endothermic reactions and mixture heating processes without external heat flux. However, the further catalyst was from the reactor inlet, the lesser it participated in mixture heating and a bigger share of stored heat was consumed due to the endothermic reactions. Thus, the highest rates of H_2 molar fractions were observed at the catalyst closest to the reformer outlet. Also, the average hydrogen mole fraction and average packed bed temperature declined from 0.377 to 0.249 and from 927 °C to 801 °C in 15 s of reforming, respectively. The significant decline in temperature and H_2 production was addressed by consequent packing heating, which increased the average packed bed temperature to the initial value. After the packed bed was heated, each catalyst temperature varied in the range of 957 °C–892 °C.

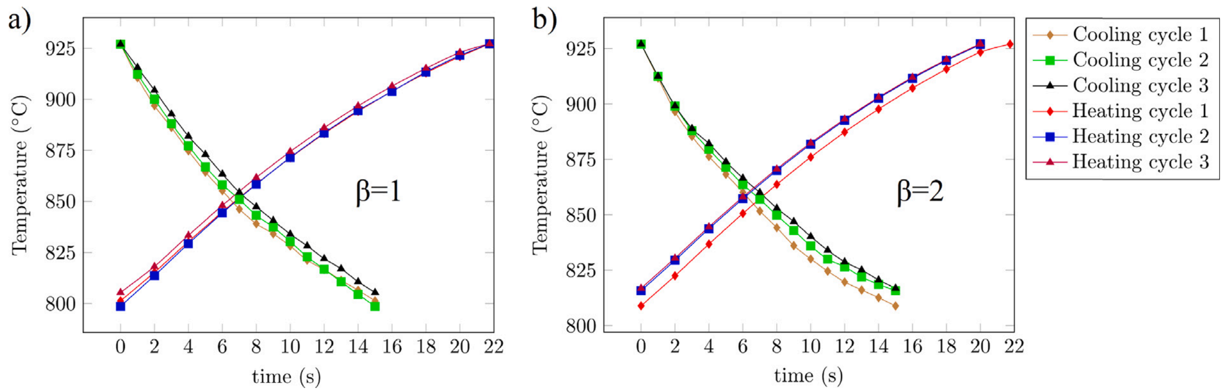

Fig. 10. Volume average temperature of all catalysts at different time on stream in three consequent cycles of packed bed cooling and heating.

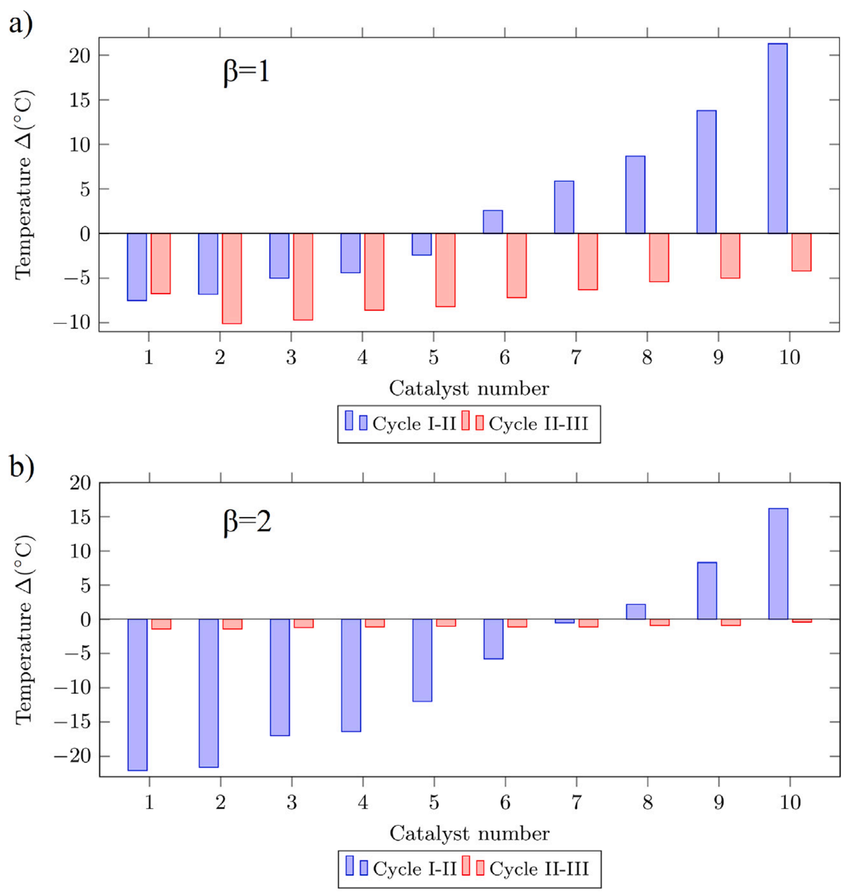

Fig. 11. The cycle-to-cycle average volume temperature difference on each catalyst for steam-to-methane ratio of 1 (a) and 2 (b) at the end of the cooling cycles.

To quantify the effect of uneven catalyst temperature distribution at the beginning of the SMR process a total of 3 consequent heating and cooling cycles were performed. The packed bed was cooled by SMR reactions for 15 s and heating to the initial average temperature took on average 21.5 s. The average inlet velocity of the steam methane mixture and flue gas was 1.5 m/s and 8 m/s, respectively. Under these conditions packed bed heating was occurring on average at a 1.5 times slower rate, than its cooling. For the cycles with $\beta = 2$ the heating in the last 2 cycles was completed after only 20 s and temperature distributions at the beginning of these cycles varied also insignificantly. The insignificant change of the parameters is a characterization of the fully developed state of the process.

CRediT authorship contribution statement

Igor Karpilov: Methodology, Writing, Visualization, Numerical modelling. **Dmitry Pashchenko:** Methodology, Writing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This work is supported by the Russian Science Foundation under grant 19-19-00327.

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